

Huda Zain Teaching Statement:

My teaching aims to help students become independent learners who can turn technical knowledge into sound engineering judgment. I encourage students to use reliable evidence, consider alternative solutions, and evaluate engineering decisions in terms of technical performance, safety, cost, scalability, reliability, and ethical implications. I want students to approach unfamiliar problems with curiosity and critical thinking, while developing the practical judgment needed to move from an initial idea to a defensible solution.

In RES 100, an introductory undergraduate research course, I help students develop this judgment through the process of conducting research. Students select a review topic that matters to them, while I guide them in turning an initial interest into a focused, evidence-based research question. I teach students to search academic databases strategically, evaluate sources, and identify relevant researchers in their field of interest. For example, when one group struggled to narrow its topic, I guided them through a targeted literature search so they could understand the existing work and define a scope they were confident pursuing. Their work culminates in a team mock poster presentation, which develops research, collaboration, and scientific communication skills.

I use assessment throughout the semester to support this progression. Quizzes and short tasks check students' understanding of threshold topics such as academic-database searching, research ethics, and the benefits, risks, and ethical considerations of AI use in research. Each assignment builds toward the final poster by asking students to select a topic, locate relevant resources, develop their research question, and begin constructing their written content. This structure allows me to provide feedback early, while students still have time to act on it and improve their work.

I also aim to make research and engineering accessible to students with different levels of prior experience. In my RES 100 section for female engineering students, I emphasise that they belong in research and engineering and that these skills can support careers in academia, industry, and policy. I extend this encouragement through virtual outreach sessions with female high-school students in Saudi Arabia, where I discuss research opportunities and pathways into engineering. For me, inclusive teaching is not only about encouraging participation; it is also about giving students the structure, confidence, and practical entry points they need to engage with technical work.

In EE311, an engineering-design course where I serve as a prototyping mentor, I help student teams turn an initial project idea into a testable engineering proposal. Rather than immediately directing students toward a particular solution, I use structured questions to prompt them to identify the intended user and decision, define what the system should output, consider where a simple version might be misleading, and select the most meaningful test or comparison. Teams share their responses with the

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class and receive feedback, helping them narrow the scope of their prototype and plan a small but convincing final demonstration. This approach reinforces an important engineering habit: defining what needs to be demonstrated before deciding how to build it.

My laboratory teaching similarly emphasises judgment through hands-on problem solving. In KIE1001 digital-systems laboratory teaching, I supported students as they progressed from basic electrical components and breadboarding to combinational and sequential logic, circuit and electronic experiments, and prototype development. I asked students to prepare by reading schematics, then checked correct assembly, safe practice, and functional operation. When a circuit did not work as expected, I guided students through systematic troubleshooting before providing an answer, using the fault as an opportunity to develop diagnostic reasoning and confidence. I want students to leave the laboratory not only knowing how a circuit should work, but also knowing how to determine why it does not.

I am prepared to teach undergraduate courses in electric circuits, electronic circuits, digital systems engineering, signals and systems, digital signal processing, control engineering, electromagnetic waves and applications, communications engineering, and electrical measurements and instrumentation. I can also contribute to associated laboratory courses, with an emphasis on safe practice, correct assembly, troubleshooting, and clear interpretation of experimental results. At advanced undergraduate and postgraduate level, my teaching interests include optical fiber communications, optics and photonics, optoelectronic devices, fiber-optic sensing, industrial instrumentation, and research methodology.

My research mentoring extends this teaching philosophy to undergraduate and Master's-level research. I support students in experimental planning, safe laboratory practice, equipment operation, troubleshooting, data analysis, and preparing research outputs. This has included mentoring a Master's researcher through the development of fiber Bragg grating sensors and optical interrogation systems, from optical-component design to laboratory execution and paper preparation; she has since progressed to PhD study at KFUPM. I have also mentored undergraduate research on LiDAR-based systems, specklegram fiber sensors, and fiber Bragg grating sensing for underwater temperature monitoring, contributing to one poster presentation, two manuscripts, and one conference submission in preparation.

Ultimately, I want students to see engineering as both a body of knowledge and a way of thinking. My aim is to help them frame meaningful questions, evaluate evidence, define appropriate tests, and make technical decisions in a logical and responsible way. Through teaching, laboratory instruction, and research mentoring, I support students in developing the judgment, practical skills, and confidence to take a problem from an initial idea to a defensible solution.